

Answer Key: Hardy-Weinberg Equilibrium and Population Genetics Practice Problems

Problem 1: Sickle Cell Trait and Malaria Resistance (Autosomal)

Given: $q = f(\beta^S) = 0.12$, $p = f(\beta^A) = 1 - 0.12 = 0.88$

a) Expected genotype frequencies under HWE:

- β^A/β^A : $p^2 = (0.88)^2 = 0.7744$
- β^A/β^S : $2pq = 2(0.88)(0.12) = 0.2112$
- β^S/β^S : $q^2 = (0.12)^2 = 0.0144$

b) Expected number of heterozygotes in population of 10,000:

$$10,000 \times 0.2112 = 2,112 \text{ individuals}$$

c) Likely violation of HWE:

The observation that β^S frequency is lower in children than in adults suggests **selection** against the β^S/β^S genotype (sickle cell anemia) before reproductive age. Children with sickle cell anemia have reduced survival, so the allele frequency appears lower in younger age groups.

Problem 2: The CCR5- Δ 32 Deletion and HIV Resistance

Given: $n_{+/+} = 720$, $n_{+/\Delta 32} = 260$, $n_{\Delta 32/\Delta 32} = 20$, $N = 1,000$

a) Observed allele frequencies:

- Number of + alleles: $2(720) + 260 = 1,440 + 260 = 1,700$
- Number of $\Delta 32$ alleles: $2(20) + 260 = 40 + 260 = 300$
- Total alleles: 2,000

$$p = f(+) = 1,700/2,000 = 0.85$$

$$q = f(\Delta 32) = 300/2,000 = 0.15$$

b) Expected genotype counts under HWE:

- Expected $+/+$: $p^2 \times N = (0.85)^2 \times 1,000 = 0.7225 \times 1,000 = 722.5$
- Expected $+\Delta 32$: $2pq \times N = 2(0.85)(0.15) \times 1,000 = 0.255 \times 1,000 = 255$
- Expected $\Delta 32/\Delta 32$: $q^2 \times N = (0.15)^2 \times 1,000 = 0.0225 \times 1,000 = 22.5$

Chi-square test:

$$\chi^2 = \sum \frac{(O - E)^2}{E} = \frac{(720 - 722.5)^2}{722.5} + \frac{(260 - 255)^2}{255} + \frac{(20 - 22.5)^2}{22.5}$$

$$= \frac{(-2.5)^2}{722.5} + \frac{(5)^2}{255} + \frac{(-2.5)^2}{22.5} = \frac{6.25}{722.5} + \frac{25}{255} + \frac{6.25}{22.5}$$

$$= 0.00865 + 0.09804 + 0.27778 = 0.38447$$

Since $0.384 < 3.84$, we **fail to reject HWE**. The population is in HWE at this locus.

c) Expected frequency of $\Delta 32/\Delta 32$ under HWE:

$q^2 = 0.0225$ or 22.5 individuals expected vs. 20 observed. The close match suggests no strong selective advantage or disadvantage for this genotype in this population.

Problem 3: X-Linked Locus: Color Blindness in Humans

Given: Frequency of color-blind males = 0.08

a) Frequency of the color-blindness allele (X^c):

For X-linked traits, males are hemizygous, so $f(\text{color-blind males}) = q = 0.08$

b) Expected frequency of color-blind females under HWE:

Color-blind females must be X^c/X^c : $q^2 = (0.08)^2 = 0.0064$ or 0.64%

c) Expected ratio of color-blind males to color-blind females:

Ratio = $q : q^2 = 0.08 : 0.0064 = 12.5 : 1$ (approximately 12.5 color-blind males for every color-blind female)

For a rare autosomal recessive disorder, the ratio of affected males to affected females is 1 : 1 (equal between sexes), whereas X-linked recessive disorders show higher incidence in males.

Problem 4: X-Linked Locus: Calico Cats

Given:

- Males: 45% orange (O/Y), 55% black (o/Y)
- Females: 36% orange (O/O), 48% calico (O/o), 16% black (o/o)

a) Allele frequency of O from male data:

$$f(O) = f(\text{orange males}) = 0.45$$

$$\text{From female data: } f(O) = f(O/O) + \frac{1}{2}f(O/o) = 0.36 + \frac{1}{2}(0.48) = 0.36 + 0.24 = 0.60$$

The frequencies are **not similar** (0.45 vs. 0.60). This suggests the population is **not in HWE** for this X-linked locus, possibly due to non-random mating, selection, or recent migration.

Problem 5: HWE as a Null Model in Conservation

Given: AA = 42, Aa = 6, aa = 2, N = 50

a) Observed allele frequencies:

- Number of A alleles: $2(42) + 6 = 84 + 6 = 90$
- Number of a alleles: $2(2) + 6 = 4 + 6 = 10$
- Total alleles: 100

$$p = f(A) = 90/100 = 0.9$$

$$q = f(a) = 10/100 = 0.1$$

b) Expected genotype counts under HWE:

- AA: $p^2 \times N = (0.9)^2 \times 50 = 0.81 \times 50 = 40.5$
- Aa: $2pq \times N = 2(0.9)(0.1) \times 50 = 0.18 \times 50 = 9$
- aa: $q^2 \times N = (0.1)^2 \times 50 = 0.01 \times 50 = 0.5$

c) Interpretation:

Observed heterozygotes (6) are much lower than expected (9). This suggests **inbreeding** or **assortative mating** in this small, isolated population, leading to a deficit of heterozygotes (Wahlund effect or inbreeding depression).

Problem 6: Overdominance: Sickle Cell Trait and Malaria

Given: $w_{AA} = 0.8$, $w_{AS} = 1.0$, $w_{SS} = 0.2$

a) Equilibrium frequency of β^S :

For overdominance, the equilibrium frequency is:

$$\hat{q} = \frac{1 - w_{AA}}{(1 - w_{AA}) + (1 - w_{SS})} = \frac{1 - 0.8}{(1 - 0.8) + (1 - 0.2)} = \frac{0.2}{0.2 + 0.8} = \frac{0.2}{1.0} = 0.2$$

Answer: $\hat{q} = 0.2$ (or 20%)

b) Effect of malaria eradication:

Without malaria, the heterozygote advantage disappears. The β^S allele becomes deleterious in homozygotes ($w_{SS} = 0.2$) with no compensating advantage in heterozygotes. Over generations, selection will reduce the frequency of β^S toward mutation-selection balance (very low frequency).

Problem 7: Overdominance in Agriculture: Cassava Mosaic Disease

Given: $w_{RR} = 0.7$, $w_{Rr} = 1.0$, $w_{rr} = 0.4$

a) Equilibrium frequency of R :

$$\hat{p} = \frac{1 - w_{rr}}{(1 - w_{RR}) + (1 - w_{rr})} = \frac{1 - 0.4}{(1 - 0.7) + (1 - 0.4)} = \frac{0.6}{0.3 + 0.6} = \frac{0.6}{0.9} = \frac{2}{3} \approx 0.667$$

$$\hat{q} = 1 - \hat{p} = 0.333$$

b) New equilibrium after virus strain change:

New fitnesses: $w_{RR} = 0.7$, $w_{Rr} = 0.6$, $w_{rr} = 0.4$

$$\hat{p} = \frac{w_{Rr} - w_{rr}}{2w_{Rr} - w_{RR} - w_{rr}} = \frac{0.6 - 0.4}{2(0.6) - 0.7 - 0.4} = \frac{0.2}{1.2 - 1.1} = \frac{0.2}{0.1} = 2$$

This yields $\hat{p} > 1$, indicating no stable internal equilibrium—the R allele will go to fixation or loss depending on initial frequency. This demonstrates the **unsustainability** of relying on a single resistance gene.

Problem 8: Underdominance: Chromosomal Rearrangements

Given: $w_{AA} = 1.0$, $w_{Aa} = 0.7$, $w_{aa} = 1.0$, $q = f(a) = 0.2$

a) Change in frequency after one generation:

Before selection (HWE):

- AA: $p^2 = 0.8^2 = 0.64$
- Aa: $2pq = 2(0.8)(0.2) = 0.32$
- aa: $q^2 = 0.2^2 = 0.04$

After selection (multiply by fitness):

- AA: $0.64 \times 1.0 = 0.64$
- Aa: $0.32 \times 0.7 = 0.224$
- aa: $0.04 \times 1.0 = 0.04$

Mean fitness: $\bar{w} = 0.64 + 0.224 + 0.04 = 0.904$

Normalized frequencies:

- AA: $0.64/0.904 = 0.70796$
- Aa: $0.224/0.904 = 0.24779$
- aa: $0.04/0.904 = 0.04425$

New $q' = f(aa) + \frac{1}{2}f(Aa) = 0.04425 + 0.123895 = 0.168145$

Answer: Frequency decreased from 0.2 to 0.168, so the a allele **decreases**.

b) At $q = 0.5$:

The underdominant equilibrium is unstable. The equilibrium frequency $\hat{q}$ is:

$$\hat{q} = \frac{w_{Aa} - w_{AA}}{2w_{Aa} - w_{AA} - w_{aa}} = \frac{0.7 - 1.0}{1.4 - 2.0} = \frac{-0.3}{-0.6} = 0.5$$

At $q = 0.5$, this is the unstable equilibrium. Any small perturbation will cause fixation of one allele.

Problem 9: Underdominance and Hybrid Zones

Given: $w_{AA} = 1.0$, $w_{Aa} = 0.6$, $w_{aa} = 1.0$

a) Unstable equilibrium frequency of a :

$$\hat{q} = \frac{w_{Aa} - w_{AA}}{2w_{Aa} - w_{AA} - w_{aa}} = \frac{0.6 - 1.0}{1.2 - 2.0} = \frac{-0.4}{-0.8} = 0.5$$

b) Population outcomes:

- At $a = 0.3$ (below equilibrium): selection favors A , which becomes fixed
- At $a = 0.7$ (above equilibrium): selection favors a , which becomes fixed

This explains hybrid zone stability—populations on either side of the zone maintain their distinct alleles due to the barrier created by unfit heterozygotes.

Problem 10: Selection and Purging of a Lethal Allele

Given: $w_{DD} = 1.0$, $w_{Dd} = 1.0$, $w_{dd} = 0.0$, $q = 0.1$

a) Genotype frequencies before selection (HWE):

- DD: $p^2 = 0.9^2 = 0.81$
- Dd: $2pq = 2(0.9)(0.1) = 0.18$
- dd: $q^2 = 0.01$

b) After selection (before normalization):

- DD: $0.81 \times 1.0 = 0.81$
- Dd: $0.18 \times 1.0 = 0.18$
- dd: $0.01 \times 0.0 = 0$

c) Mean fitness and normalization:

$$\bar{w} = 0.81 + 0.18 + 0 = 0.99$$

Normalized frequencies:

- DD: $0.81/0.99 = 0.81818$
- Dd: $0.18/0.99 = 0.18182$
- dd: 0

d) Frequency of d in next generation:

$$q' = f(dd) + \frac{1}{2}f(Dd) = 0 + \frac{1}{2}(0.18182) = 0.09091$$

e) Will d be completely eliminated?

No. While selection removes the lethal dd homozygotes each generation, the d allele is "hidden" in heterozygotes. The frequency decreases each generation but approaches zero asymptotically; it never reaches exactly zero unless drift or mutation counteracts selection.

Problem 11: Unidirectional Mutation: Congenital Hypertrichosis

Given: $\mu = 1 \times 10^{-5}$, $s = 1 - 0.7 = 0.3$ (for dominant deleterious allele)

a) Equilibrium frequency under mutation-selection balance:

$$\text{For a dominant deleterious allele: } \hat{q} = \frac{\mu}{s} = \frac{1 \times 10^{-5}}{0.3} = 3.33 \times 10^{-5}$$

b) Comparison with recessive allele:

For a recessive deleterious allele with same μ and s , $\hat{q} = \sqrt{\frac{\mu}{s}} = \sqrt{3.33 \times 10^{-5}} \approx 0.00577$, which is much larger.

Dominant deleterious alleles are exposed to selection in heterozygotes, so they are removed more efficiently and reach much lower equilibrium frequencies than recessive alleles.

Problem 12: Recessive Lethal: Tay-Sachs Disease

Given: Incidence in general population = $1/320,000$, in Ashkenazi = $1/3,600$, $s = 1$

a) Allele frequency:

$$q^2 = \text{incidence}$$

$$\text{General: } q = \sqrt{1/320,000} = \sqrt{3.125 \times 10^{-6}} = 0.00177 \text{ (about 0.18\%)}$$

$$\text{Ashkenazi: } q = \sqrt{1/3,600} = \sqrt{2.78 \times 10^{-4}} = 0.01667 \text{ (about 1.67\%)}$$

b) Estimated mutation rate:

Under mutation-selection balance for recessive lethal: $q = \sqrt{\mu/s} = \sqrt{\mu}$ (since $s = 1$)
 $\mu = q^2 = 1/3,600 = 2.78 \times 10^{-4}$

c) Evolutionary force:

The estimated mutation rate is much higher than actual mutation rates ($\sim 10^{-8}$ to 10^{-6}). The high frequency is likely due to **heterozygote advantage** (overdominance), possibly conferring resistance to tuberculosis or other infectious diseases in Ashkenazi Jewish populations.

Problem 13: Bidirectional Mutation and Equilibrium

Given: $\mu_{A \rightarrow a} = 1 \times 10^{-5}$, $\mu_{a \rightarrow A} = 5 \times 10^{-6}$

a) Equilibrium frequency:

$$\hat{q} = \frac{\mu_{A \rightarrow a}}{\mu_{A \rightarrow a} + \mu_{a \rightarrow A}} = \frac{1 \times 10^{-5}}{1 \times 10^{-5} + 5 \times 10^{-6}} = \frac{1 \times 10^{-5}}{1.5 \times 10^{-5}} = \frac{2}{3} \approx 0.667$$

b) Will q reach 0.5?

Starting from $q = 0$, the frequency increases toward equilibrium at 0.667. Since $0.5 < 0.667$, the frequency will **exceed 0.5** on its way to equilibrium. However, with only mutation (no selection, no drift), the approach is deterministic and monotonic.

Problem 14: Mutation-Selection Balance in Antibiotic Resistance

Given: $s = 0.2$ (fitness cost of resistance in absence of antibiotics), $\mu = 1 \times 10^{-8}$, recessive resistance

a) Equilibrium frequency:

For recessive deleterious allele: $\hat{q} = \sqrt{\frac{\mu}{s}} = \sqrt{\frac{1 \times 10^{-8}}{0.2}} = \sqrt{5 \times 10^{-8}} = 2.24 \times 10^{-4}$

b) During treatment:

When rifampicin is present, the selection coefficient reverses—resistant bacteria have higher fitness. The equilibrium shifts dramatically, and the resistant allele frequency increases toward 1. If treatment is stopped prematurely, the resistant population may have increased sufficiently to cause relapse, and without the drug, the fitness cost may slowly reduce resistance frequency, but the infection may already be established.

Problem 15: Mutation-Selection Balance in a Conservation Context

Given: $N = 500$, $\mu = 1 \times 10^{-6}$, $s = 0.5$

a) Expected equilibrium frequency:

$$q = \sqrt{\frac{\mu}{s}} = \sqrt{\frac{1 \times 10^{-6}}{0.5}} = \sqrt{2 \times 10^{-6}} = 0.001414$$

b) Deviation from equilibrium:

In a small population ($N = 500$), **genetic drift** will cause deviations from the deterministic equilibrium. The actual frequency may fluctuate around this value, and the allele may be lost or fixed by chance, especially if selection is weak relative to drift ($N_e s < 1$).

Problem 16: Gene Flow and the Sickle Cell Allele

Given: $q_m = 0.12$, $q_0 = 0.06$, $w_{AA} = 1.0$, $w_{AS} = 1.0$, $w_{SS} = 0.2$

a) Frequency after migration:

$$q'_{\text{migration}} = (1 - m)q_0 + mq_m$$

b) Frequency after selection:

After migration and HWE, frequencies are:

- AA: p^2 , AS: $2pq$, SS: q^2

After selection:

$$\begin{aligned}\bar{w} &= p^2(1) + 2pq(1) + q^2(0.2) = 1 - 0.8q^2 \\ q_{\text{after selection}} &= \frac{q^2(0.2) + pq(1)}{\bar{w}} = \frac{q^2(0.2) + q(1 - q)}{\bar{w}}\end{aligned}$$

c) Estimate migration rate at equilibrium:

At equilibrium, the decrease due to selection equals the increase from migration. For small q , $q_{\text{after selection}} \approx q - 0.8q^2$ and migration adds $m(q_m - q)$. Setting $q_{\text{after}} = q_0$ and solving yields $m \approx 0.03$ (approximately 3% migration per generation).

Problem 17: Gene Flow and Pesticide Resistance in Crop Pests

Given: $q_m = 0$, $q_0 = 0.8$, $m = 0.1$, $w_{RR} = 1.0$, $w_{Rr} = 1.0$, $w_{rr} = 0.6$

a) Frequency after migration:

$$q' = (1 - m)q_0 + mq_m = (0.9)(0.8) + (0.1)(0) = 0.72$$

b) Refuge strategy explanation:

The refuge (unsprayed area) maintains susceptible individuals (rr). Migration from the refuge introduces susceptible alleles into the cotton field, reducing the frequency of resistance alleles. Even though resistance is advantageous in the field, gene flow from the refuge slows adaptation and delays the evolution of resistance, preserving pesticide efficacy.

Problem 18: Counteracting Gene Flow: Conservation of an Endangered Plant

Given: $q_0 = 0.1$, $q_m = 1.0$, $m = 0.05$, $w_{A_1A_1} = 1.0$, $w_{A_1A_2} = 0.8$, $w_{A_2A_2} = 0.6$

a) Frequency after migration:

$$q_{\text{after migration}} = (1 - m)q_0 + mq_m = (0.95)(0.1) + (0.05)(1.0) = 0.095 + 0.05 = 0.145$$

b) Frequency after selection:

Before selection (HWE with $q = 0.145$, $p = 0.855$):

- A_1A_1 : $p^2 = 0.731025$
- A_1A_2 : $2pq = 2(0.855)(0.145) = 0.24795$
- A_2A_2 : $q^2 = 0.021025$

After selection (multiply by fitnesses):

- A_1A_1 : $0.731025 \times 1.0 = 0.731025$
- A_1A_2 : $0.24795 \times 0.8 = 0.19836$
- A_2A_2 : $0.021025 \times 0.6 = 0.012615$

Mean fitness: $\bar{w} = 0.731025 + 0.19836 + 0.012615 = 0.942$

Normalized:

- A_1A_1 : $0.731025/0.942 = 0.7760$
- A_1A_2 : $0.19836/0.942 = 0.2106$
- A_2A_2 : $0.012615/0.942 = 0.0134$

$$q' = f(A_2A_2) + \frac{1}{2}f(A_1A_2) = 0.0134 + 0.1053 = 0.1187$$

c) Equilibrium frequency:

At equilibrium, migration and selection balance. The equilibrium is approximately $\hat{q} \approx 0.125$

d) Conservation action:

Create physical barriers (buffer zones) between the endangered and weedy species, remove weedy individuals near the population, or establish new populations in areas without the weedy species to reduce hybridization and gene flow.

Problem 19: Island-Continent Model for an X-Linked Locus

Given: $q_m = 0.01$ (continent), $q_0 = 0$ (island), $m = 0.02$ (migration rate for males)

After one generation of migration:

Island males: Males inherit their X from their mothers. After migration, the male population consists of $(1 - m)$ island-born males and m immigrant males. Island-born males have frequency q_{female} from the previous generation. Initially $q_{female} = 0$, so:

$$q'_{males} = (1 - m)q_{females} + mq_m = (0.98)(0) + (0.02)(0.01) = 0.0002$$

Island females: Females inherit one X from father and one from mother. Before migration, both parents had $q = 0$. After migration:

- Father's contribution: $q'_{males} = 0.0002$
- Mother's contribution: $q_{females} = 0$ (mothers are island females before migration)

$$q'_{females} = \frac{q'_{males} + q_{females}}{2} = \frac{0.0002 + 0}{2} = 0.0001$$

Problem 20: Asymmetric Gene Flow in a Hybrid Zone

Given: $w_{AA} = 1$, $w_{Aa} = 0.7$, $w_{aa} = 1$, $m_A = 0.05$ (from A to a), $m_a = 0.01$ (from a to A)

a) Species range expansion:

More individuals move from A to a ($m_A = 0.05$) than from a to A ($m_a = 0.01$). This asymmetric gene flow favors the expansion of species A into a territory.

b) Interaction between migration and underdominance:

Underdominance creates a barrier to gene flow—hybrids have reduced fitness. Asymmetric migration biases this barrier. The hybrid zone may shift in the direction of greater migration pressure. The zone's stability depends on the balance between selection against hybrids and the rate of immigration. Strong selection can maintain a narrow zone despite asymmetry, but extreme asymmetry may cause the zone to move as one species gradually replaces the other.